National Aeronautics and Space Administration

Cross-mission expiMap analysis recovers established tissue responses and identifies complementary pathway shifts in mouse spaceflight transcriptomes

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ABSTRACT

Mission differences can obscure spaceflight responses shared across studies.
We mapped NASA OSDR mouse bulk RNA sequencing samples to tissue-matched ARCHS4 references.
Reactome-constrained expiMap models analyzed about 2,000 highly variable genes.
Flight-ground shifts were balanced across projects and evaluated with gene-set enrichment,
project holdouts, three complete training runs, composition proxies, and member-gene review.
Literature review was performed after scoring. Retained programs were lower for thymic repair,
skin maintenance, liver adaptive immunity, and splenic adaptive and innate immunity.
Spleen showed the strongest result across multiple pathways.

INTRODUCTION

PROJECT OBJECTIVE

Identify gene programs that shift consistently across spaceflight missions.

THE CENTRAL CHALLENGE

Observed expression | Spaceflight biology + Tissue biology + Animal variation + Study / protocol

Study identity and protocol can dominate an unconstrained expression representation.
Tissue-matched reference mapping, accession conditioning, and equal project weighting
reduce this bias without claiming to remove all mission confounding.

FINAL ANALYSIS SCOPE

Tissue	ARCHS4	OSDR	Projects	Programs
Thymus	1,362	117	5	387
Skin	2,593	151	4	319
Liver	5,000	197	9	364
Spleen	6,289	100	5	360

OSDR counts are primary analysis samples.
Spleen excludes one condition-strain-confounded project.

ROBUSTNESS GATE

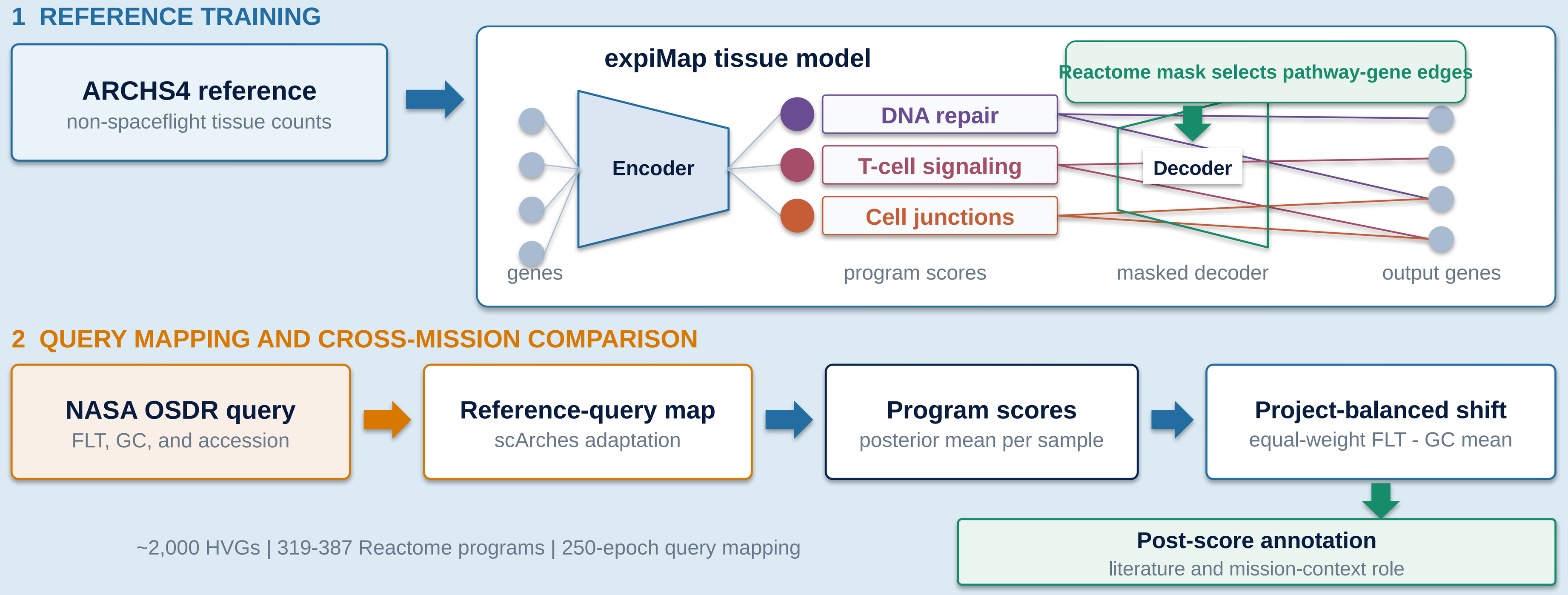
- ssGSEA + GSEA
- Held-out projects
- 3 full trainings
- Composition proxies
- Member genes
- Literature review

Complementary programs add a plausible perspective beyond the dominant phenotype
in prior literature. De novo nodes were not retained in the final models.

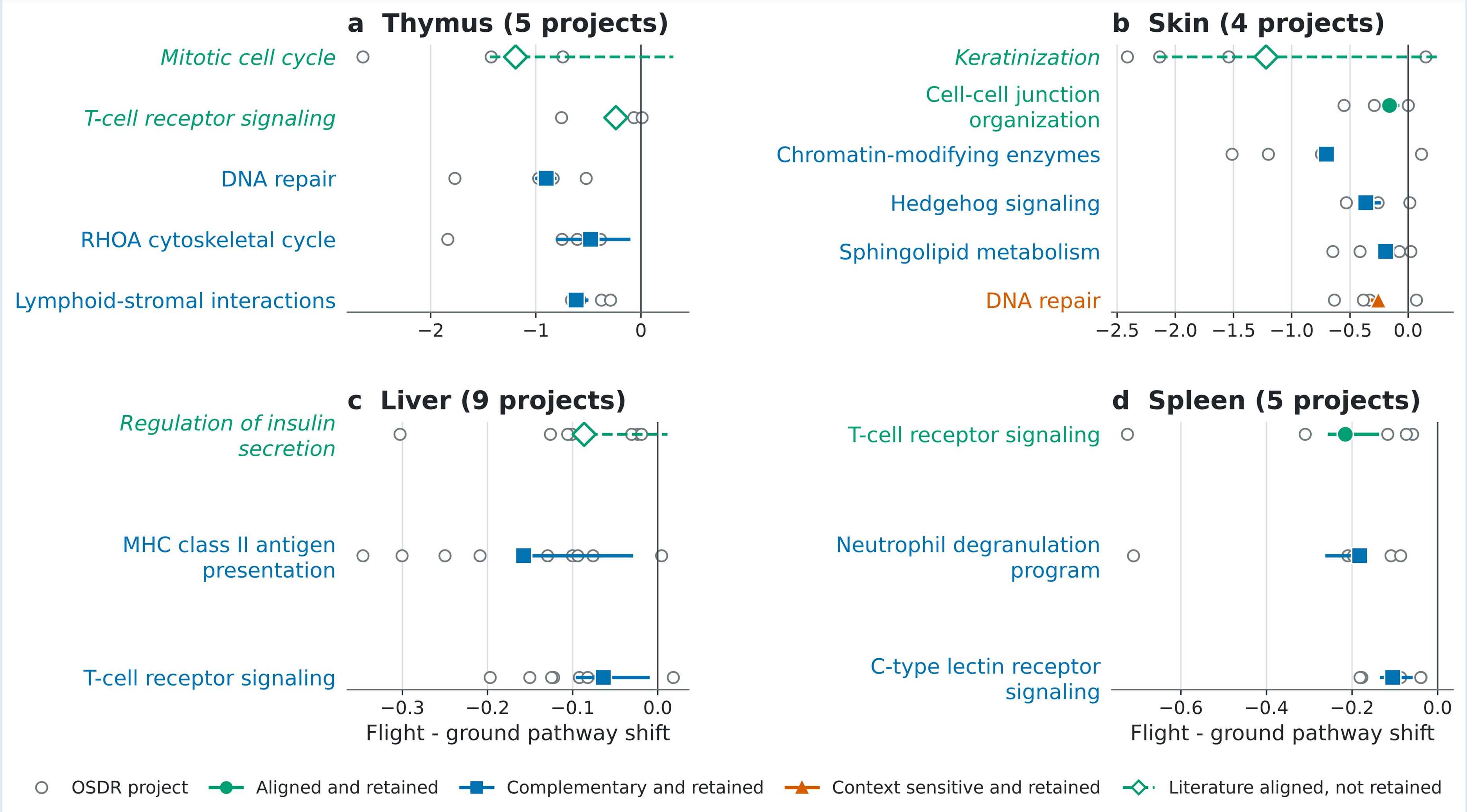
METHODS

Pathway structure is wired into the decoder

Train a tissue reference, then map flight and ground samples into the same annotated program space.



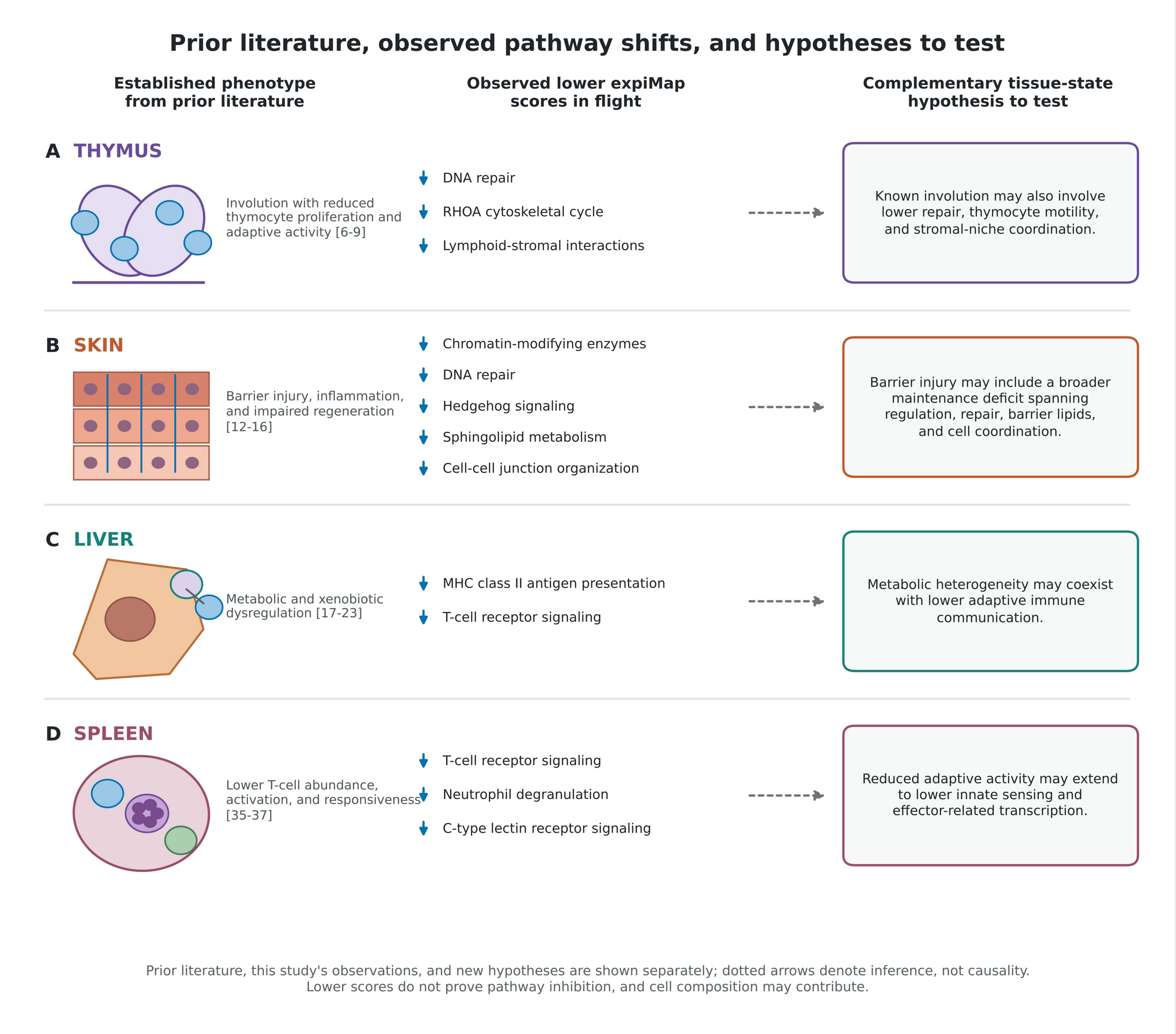
RESULTS: ALIGNED ANCHORS AND RETAINED PROGRAM SHIFTS



Filled markers show 13 retained programs. Open diamonds are literature-aligned context only, not retained findings; keratinization reversed in one of three trainings. Colored ranges span three complete trainings; axes are tissue specific.

RESULTS: BIOLOGICAL INTERPRETATION

Prior evidence, observed shifts, and hypotheses to test



CONCLUSIONS

- Spleen was strongest: T cell receptor, neutrophil degranulation, and C-type lectin receptor programs were lower across five projects and three trainings; all had GSEA FDR <0.05.
- Skin and thymus emphasized lower tissue maintenance programs; liver showed lower adaptive immune signaling amid heterogeneous metabolism.
- Bulk latent scores are testable hypotheses, not causal or cell-resolved measures of pathway activity.

REFERENCES

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github.com/jasont314/nasa-mouse